

Hardware-Efficient Optimal Error-Priority Control for Series-End VSI with ZSC Suppression

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Abstract—This paper presents a novel Optimal Error-Priority Control (OEP) strategy for Series-End VSI, achieving high performance with minimal computational overhead. The OEP approach integrates zero-sequence current (ZSC) suppression in ABC coordinates using a compact one-dimensional lookup table, eliminating the need for complex real-time calculations of three-dimensional space vector modulation. The direct lookup phase-wise current control avoids additional linear compensators and reduces switching losses by utilizing variable switching frequency to cut switching events per cycle. OEP ensures minimal execution time by facilitating simple hardware implementation on standard microcontrollers with optional programmable logic arrays for even higher computational efficiency at a reduced cost. Simulations and C-HIL experiments validate the approach with RL loads and PMSM drives with prominent third harmonic flux components, demonstrating sub-2 μ s execution time on a TMS320F28335 target. Confirmed superior current quality with maximal DC bus utilization and adaptability to various configurations underscores the industrial viability of the proposed OEP framework.

Index Terms—PMSM, Series-end Voltage source inverter (SE-VSI), Zero-sequence current (ZSC), lookup table-based control, field-oriented control (FOC).

I. INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs), particularly open-winding (OW) PMSMs, are gaining more attention due to their high efficiency, high DC-link utilization, improved fault tolerance, and enhanced control flexibility [1] – [8]. However, OW-PMSM drives face challenges like increased converter complexity and Zero-Sequence Current (ZSC) mitigation issues [1] – [6]. To overcome these, researchers have explored advanced inverter topologies, with series-end voltage source inverters (SE-VSIs) emerging as an up-and-coming alternative to double-VSI-based configurations [2], offering comparable DC-link voltage utilization and fault tolerance but requiring fewer switching devices, thus reducing cost and complexity [3] – [9]. Despite their advantages, SE-VSI-fed PMSM drives still face a significant challenge when it comes to ZSC and Common Mode Voltage (CMV) suppression [3] – [9]. Uncontrolled ZSC degrades performance by increasing harmonic distortion, machine losses, torque ripple, and electromagnetic interference (EMI) [2], [6]. ZSC arises from modulation techniques, inverter nonlinearity (e.g., deadtimes), and harmonic components in the back-EMF [1], [4] – [7], [10].

Various ZSC suppression methodologies have been proposed for SE-VSI-fed PMSM drives. Early carrier-based methods, such as constant CMV PWM [3], limit CMV amplitude to one-third of traditional schemes, reducing leakage current and EMI. However, they introduce current switching harmonics, retain

residual CMV, and fail to eliminate common-mode voltage spikes from dead zone effects [3]. Model Predictive Current Control (MPCC) strategies offer improved performance [4], [5]. Multivector MPCC enhances steady-state behavior using multiple voltage vectors [4], while active MPCC reduces ZSC ripple [5]. However, both methods are sensitive to model parameters, leading to control delays [4], [5]. More advanced solutions, such as Three-Dimensional Space Vector Modulation (3D-SVM) [6], synthesize voltages in the $\alpha\beta\gamma$ -space, expanding the modulation range and enabling intuitive overmodulation adjustment. However, they require complex sector identification, duty cycle computation, and vector averaging dwell times, introducing delays. Additionally, the sub-tetrahedron distribution in 3D space lacks symmetry, thus complicating reference vector determination [6]. Recent advancements include improved PWM strategies that eliminate complex sector identification and incorporate deadtime compensation using Adaline Neural Networks (ANN) for harmonic extraction [7]. While high-performance techniques such as MPCC, 3D-SVM, and ANN-based PWM enhance control, they impose heavy computational demands, making them impractical for real-time, low-cost industrial applications [4] – [8], [11]. Many modulation-based strategies like advanced PWM [7] and 3D-SVM [6] struggle to suppress ZSC in non-sinusoidal back-EMF PMSMs, where third-harmonic components further complicate control [2], [4], [5]. Such PWM-based approaches also increase switching losses, causing higher thermal stress and EMI [4], [5], [12], [13]. Moreover, conventional OW-PMSM drive solutions rely on separate linear compensators, such as the proportion integration resonance (PIR) compensator [14], [15] and 3D-SVM for proper ZSC suppression. While practical, such an approach adds significant complexity and increases computational burden.

In contrast, the proposed OEP scheme inherently addresses ZSC within its predictive phase-wise control framework, obviating the need for any additional linear compensators or dwell times calculations. By eliminating this extra control loop and specialized modulation, OEP simplifies the implementation, enhances dynamic performance, and reduces real-time computation by leveraging a precomputed 1D lookup table (LUT). This ensures superior computational efficiency, thus offering a clear potential advantage over existing methods.

This paper proposes a novel Optimal Error Priority Control (OEP) strategy for SE-VSI-fed PMSM drives, achieving direct ZSC suppression in ABC coordinates without modulation or reference averaging. Unlike conventional 3D space vector synthesis or duty cycle calculations, OEP utilizes a precomputed LUT for real-time optimal switching state selection, enabling phase-wise differential and ZSC error

compensation. The method offers a hardware-efficient alternative to MPCC and 3D-SVM, maximizing DC bus utilization while minimizing computational burden.

Key Contributions of OEPC include:

1. **A novel framework** for direct ZSC suppression in ABC coordinates using a compact 1D-LUT eliminates duty cycle averaging and additional linear compensators, achieving minimal ZSC in real-world conditions and verifying absolute zero ZSC in ideal conditions.

2. **Computational efficiency** with a sub-1 μ s execution time on the TMS320F28335 DSP, making it real-time compatible. The LUT-based approach eliminates iterative trigonometric calculations, reducing processing overhead.

3. **Variable-frequency on-off control**, dynamically adjusting switching frequency (f_{sw}) to reduce switching losses by up to ~30%, optimizing EMI performance.

4. **Comprehensive validation** through digital simulations and C-HIL experiments for target hardware verification, demonstrating superior ZSC suppression and minimal current THD across f_{sw} of 5–20 kHz, particularly in non-sinusoidal back-EMF PMSMs, even for drives with significant 3rd harmonic to fundamental flux ratios.

OEPC's adaptability could extend to other motor drive configurations, including polyphase SE-VSI-fed PMSMs and induction motors, underscoring its industrial relevance as a high-performance, computationally efficient alternative.

The remaining paper is structured as follows: **Section II** covers the theoretical background, including PMSM modeling and control, the SE-VSI topology and switching states, and the ZSC suppression challenges. **Section III** details the OEPC methodology, error-priority selection, and LUT implementation. **Section IV** presents validation results, including simulations, C-HIL experiments, and comparisons with state-of-the-art methods. **Section V** concludes with key findings and future research directions.

II. BACKGROUND AND FORMULATIONS

A. OW-PMSM Model with ZSC Considerations

ZSC in OW-PMSMs arises primarily from two mechanisms: (1) inverter-side CMV resulting from sub-optimal modulation sequences and VSI nonlinearities, such as dead time and device voltage drops [2], [4]; and (2) motor-side CMV due to third-harmonic rotor flux, which leads to non-sinusoidal back-EMF, particularly in salient pole interior PMSMs [11], [15]. Additionally, resolver excitation frequencies of some speed encoders may also contribute to ZSC [1]. To model the OW-PMSM behavior under SE-VSI operation while considering ZSC effects, the voltage behind reactance (VBR) PMSM model [16] was enhanced to incorporate 3rd harmonic flux effects [2], [17], resulting in the following governing equations:

$$\begin{cases} \lambda_d = L_d i_d + \lambda_m, & \lambda_q = L_q i_q \\ \lambda_0 = L_0 i_0 + K_m \lambda_m \cos(3\omega_e t) \end{cases} \quad (1)$$

$$\begin{cases} v_d \approx r_s i_d + e_d \\ v_q \approx r_s i_q + e_q \\ v_0 \approx r_s i_0 + L_0 \frac{di_0}{dt} + e_0 \end{cases} \quad (2)$$

$$\begin{cases} e_d \approx (L_d - L_q) \omega_e i_q \\ e_q \approx (L_d - L_q) \omega_e i_d + \omega_e \lambda_m \\ e_0 \approx -3K_m \lambda_m \omega_e \sin(3\omega_e t) \end{cases} \quad (3)$$

$$J \frac{d\omega_m}{dt} + F \omega_m = T_e - T_m \quad (4)$$

$$T_e = \frac{3p}{2} \left((L_d - L_q) i_d i_q + \lambda_m i_q \right) - 9p K_m \lambda_m i_0 \sin(3\theta_e), \quad (5)$$

where λ, v, e, i , and T represent flux linkage, voltage, back EMF, current, and torque variables, respectively, ω_m and $\omega_e = p\omega_m = \theta_e/t$ denote the mechanical and electrical angular frequencies, respectively, with p denoting the number of pole pairs, while r_s, L_d, L_q , and L_0 denote the stator resistance and rotating reference frame (RRF) inductances, respectively. The magnitude of the permanent magnet (PM) fundamental flux component is denoted as λ_m , while K_m denotes the 3rd harmonic to fundamental PM flux ratio, respectively. The 3rd harmonic back EMF component contributes to the unwanted ZSC:

$$i_0 = \frac{i_a + i_b + i_c}{3}. \quad (6)$$

As per the VBR concept, the steady state back EMF expressions are obtained by first transforming the RRF currents into their stationary reference frame (SRF): $i_d, i_q = I_m \cos(\omega_e t) \pm I_m \sin(\omega_e t)$, to allow for phasor-based derivative approximations, vis, $di_d/dt \approx +\omega_e i_q$ and $di_q/dt \approx -\omega_e i_d$.

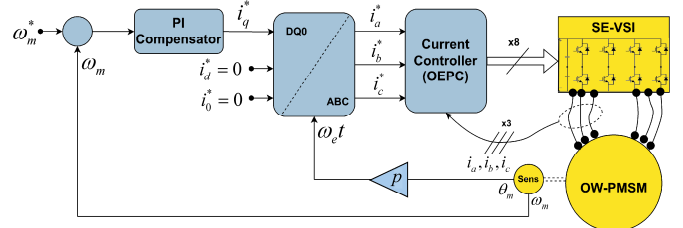


Fig. 1. Block Diagram of OW-PMSM FOC-based Speed Control with ABC Coordinates Current controller (OEPC).

B. OW-PMSM Field Oriented Control

The OW-PMSM Field-Oriented Control (FOC) scheme employed in this work consists of a PI-based speed controller and a current controller operating in the SRF. Fig. 1 presents the full control block diagram. The PI controller determines the q-axis current reference from the speed error while maintaining d-axis and zero-axis references at zero. Using the symmetric optimum method [18], the PI gains were set as:

$$K_p = \frac{2J\omega_c^2}{3p\lambda_m}, \quad K_i = \frac{4Jp\omega_c - 2F}{3p\lambda_m}, \quad (7)$$

where ω_c is the desired speed loop bandwidth, J is the moment of inertia, and F is the viscous friction coefficient.

C. The Series-End VSI and ZSC Suppression

The SE-VSI topology, illustrated in Fig. 2, consists of four independent half-bridge legs (L1 to L4), where each adjacent legs govern one phase load connection. Unlike traditional VSI, SE-VSI introduces new zero-sequence current (ZSC) paths due

to the absence of a neutral point. The key challenge is that certain Switching States (SS) inherently induce circulating currents through antiparallel diodes, preventing ideal ZSC suppression. Since switching off both leg transistors results in loss of current control, only two valid SSs are assigned to each leg, resulting in a total of $2^4 = 16$ SSs, where each SS determines the voltage applied to the load phases. To analyze these effects, Table I presents the 16 SSs, detailing the impact of each SS on the three-phase currents, i_a, i_b, i_c , and on the ZSC, i_0 (6), using the '+' and '-' prefixes to indicate positive (↑) or negative (↓) impact, respectively. Unlike in 3D-SVM, where they may be utilized for vector averaging [2], [6], the zero vectors (#0 and #15) do not influence the phase currents. They are avoided in OEPC, reducing excursion times and improving performance. Notably, 8 out of the 16 SSs impact the ZSC, while 8 SSs impact each of the three-phase currents (with overlaps). This distribution creates a complex control scenario where multiple objectives must be balanced simultaneously.

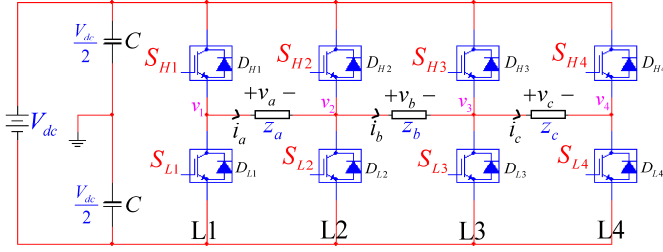


Fig. 2. Three-Phase SE-End-VSI with Open-Winding Load.

#	SS($L_1L_2L_3L_4$)	Impact
0	0000	N/A
1	0001	$-i_c, -i_0$
2	0010	$-i_b, +i_c$
3	0011	$-i_b, -i_0$
4	0100	$-i_a, +i_b$
5	0101	$-i_a, +i_b, -i_c, -i_0$
6	0110	$-i_a, +i_c$
7	0111	$-i_a, -i_0$
8	1000	$+i_a, +i_0$
9	1001	$+i_a, -i_c$
10	1010	$+i_a, -i_b, +i_c, +i_0$
11	1011	$+i_a, -i_b$
12	1100	$+i_b, +i_0$
13	1101	$+i_b, -i_c$
14	1110	$+i_c, +i_0$
15	1111	N/A

The ZSC component flows along a path that consists of the VSI's CMV, defined as the outer legs' voltage difference, $v_1 - v_4$ (c.f. Fig. 2), and a pseudo-series connection of the load impedances [6, Fig. 2b]. Many existing SE-VSI control schemes utilize this concept to suppress ZSC by controlling the SE-VSI's CMV [2], [3]. However, even if both transistors of the two middle legs, L2 and L3, are turned off simultaneously, due to the SE-VSI topology and the antiparallel diodes, the load impedances cannot form series connections under any chosen SS, as such would violate KCL. Instead, when specific SS

occur, the antiparallel diodes conduct, forcing the load impedances to either form parallel paths with the DC source or establish self-circulating current loops, thus maintaining current continuity within the load phases. Moreover, as verified by comprehensive simulations, when a ZSC-impacting SS is activated, due to the antiparallel diodes action and the inherent phase interdependence of the middle legs, either only one load impedance would connect in parallel with the DC source or two load impedances would connect in antiparallel (opposing polarities), while the remaining impedances would self-circulate. Therefore, the net ZSC impact would remain limited to only a 1/3-fold. This means that the SE-VSI can only control the ZSC by controlling only a single-phase current at a time. This inherent ZSC control limitation forms the foundation of the proposed OEPC framework, which leverages this characteristic to achieve optimal phase current control with seamless phase-wise ZSC suppression.

III. OPTIMAL ERROR-PRIORITY CONTROL (OEPC)

The proposed OEPC framework addresses the "one-current-at-a-time" ZSC control limitation by employing a 1D-LUT for optimal SS selection, integrating three-phase control and ZSC suppression into a seamless control process by utilizing error priority-based criteria to optimize the phase-wise current control. This efficient on-off-based current control results in a variable f_{sw} , bounded by an adjustable upper limit of f_{max} , which is set by the sampling period, T_s , and given by the following relation [19], [20]:

$$f_{max} = \frac{1}{2T_s}. \quad (8)$$

Implemented directly in the SRF, the OEPC process eliminates the need for 3D-SVM averaging and additional linear compensators while ensuring minimal switching losses through its efficient self-adjusted f_{sw} , which varies between $0.5f_{max}$ and f_{max} , with an average of $0.75f_{max}$.

A. Differential Mode and Common Mode Current Errors

With the employed phase-wise current regulation, each phase current ($i_x, x \in \{a, b, c\}$) can correct either a differential-mode (DM) error (for torque control) or a common-mode (CM) error (for ZSC suppression). This yields six independent errors: three DM and three CM errors (one per phase). The DM reference currents exclude the desired ZSC reference, i_0^* , which is ideally zero. Namely, for phase x , with complimentary phases y, z (e.g., $x = a \Rightarrow y = b, z = c$), there is:

$$i_{DM(x)}^* = i_x^* - i_0^* = \frac{2i_x^* - i_y^* - i_z^*}{3}, \text{ where } i_0^* = \frac{i_x^* + i_y^* + i_z^*}{3},$$

and the three DM tracking errors become:

$$\varepsilon_{DM(a)} = i_{DM(a)}^* - i_x = \frac{2i_x^* - i_y^* - i_z^*}{3} - i_x. \quad (9)$$

By equating i_0^* with i_0 (6), then solving for i_x , there is: $i_x = 3i_0^* - (i_y + i_z)$, where the right side of the equality serves as the reference value. As a result, the three CM errors become:

$$\varepsilon_{CM(x)} = \frac{3i_0^* - (i_y + i_z) - i_x}{3}. \quad (10)$$

A 1/3 scaling factor was applied to balance the correction priority, avoiding the CM error corrections from disproportionately overshadowing the DM current regulation.

B. Error-Priority Selection and Reduction

With two error types per phase, where each phase current can be governed to regulate one error type at a time (DM or CM), an error-type priority selection process must be implemented by applying a pair-wise error comparison, in which the absolute values of DM and CM errors are compared per-phase. This way, the most critical phase-wise error is selected, thus reducing the error set from six to three (one critical per phase), vis,

$$\varepsilon_{crit(x)} = \varepsilon_{DM(x)} \text{ if } |\varepsilon_{DM(x)}| \geq |\varepsilon_{CM(x)}| \text{ else } \varepsilon_{CM(x)}. \quad (11)$$

Additionally, an error mode predominance (EMP) flag, F_{mp} , is set to 0 if the majority of the phase-wise critical errors (CE) are of CM type. Else, it is set to 1, vis,

$$F_{mp} = \begin{cases} 1, & \text{if 2 } \varepsilon_{crit(x)} \text{ are of DM type} \\ 0, & \text{if 2 } \varepsilon_{crit(x)} \text{ are of CM type} \end{cases} \quad (12)$$

Next, the three selected phase-wise CEs, $\varepsilon_{crit(x)}$, are sorted in descending order of absolute values, while their order is encoded as a 3-bit error priority code (EPC) defining which phase-wise error is maximum (highest priority), intermediate (medium priority), and minimum (lowest priority). The EPC, which consists of three binary variables, P_{ab} , P_{bc} , and P_{ca} , is obtained using three logical comparisons,

$$\begin{cases} P_{ab} = [|\varepsilon_{crit(a)}| > |\varepsilon_{crit(b)}|] \\ P_{bc} = [|\varepsilon_{crit(b)}| > |\varepsilon_{crit(c)}|] \\ P_{ca} = [|\varepsilon_{crit(c)}| > |\varepsilon_{crit(a)}|], \end{cases} \quad (13)$$

where $[]$ is the Iverson brackets notation. The error priority orders (EPO) and their corresponding EPCs are listed in Table II, where an EPO of CBA (EPC = 001) means that $\varepsilon_{crit(c)}$ is the most critical (the prime error), followed by phases B and A, namely, $|\varepsilon_{crit(c)}| > |\varepsilon_{crit(b)}| > |\varepsilon_{crit(a)}|$. Note that 000 and 111 are unused since only $3!=6$ unique permutations exist. In addition to the EPCs, the sorting process also generates a 3-bit error signs code (ESC), indicating each CE's sign using a second set of binary variables, S_{mx} , S_{md} , and S_{mn} , corresponding to the highest priority, medium priority, and least priority errors, respectively, where a value of 0 (negative) means that the current must be decreased and a values of or 1 (positive) means that the current must be increased.

TABLE II
ERROR PRIORITY CODE AND ORDER

#	EPC($P_{ab}P_{bc}P_{ca}$)	EPO
0	000	N/A
1	001	CBA
2	010	BAC
3	011	BCA
4	100	ACB
5	101	CAB
6	110	ABC
7	111	N/A

For example, if EPC = 001 (CBA priority) and ESC = 101, then the most CE, $\varepsilon_{crit(c)}$, is positive, the intermediate CE, $\varepsilon_{crit(b)}$,

is negative, and the least CE, $\varepsilon_{crit(a)}$, is positive. The three ESC bits can be logically expressed as follows:

$$\begin{aligned} \text{with: } S_x &= [\varepsilon_{crit(x)} > 0], \text{ and } x \in \{a, b, c\}, \\ \begin{cases} S_{mx} = S_a \overline{P_{ca}} P_{ab} + S_b \overline{P_{ab}} P_{bc} + S_c \overline{P_{bc}} P_{ca} \\ S_{md} = S_a (\overline{P_{ca}} \oplus \overline{P_{ab}}) + S_b (\overline{P_{ab}} \oplus \overline{P_{bc}}) + S_c (\overline{P_{bc}} \oplus \overline{P_{ca}}) \\ S_{mn} = S_a P_{ca} \overline{P_{ab}} + S_b P_{ab} \overline{P_{bc}} + S_c P_{bc} \overline{P_{ca}}. \end{cases} \end{aligned} \quad (14)$$

The seven key binary variables (7-bit LUT input address),

$$[P_{ab}, P_{bc}, P_{ca}, S_{mx}, S_{md}, S_{mn}, F_{mp}], \quad (15)$$

are encoded into a 7-bit decimal index. The 1D-LUT then utilizes this index to determine the optimal SSs for the next processing step.

C. Switching State Selection (LUT Programming)

The core of OEPC lies in its comprehensive decision-making, which follows a structured, precomputed mapping rather than a runtime-based, computationally intensive algorithm for the optimal selection of SSs. This is achieved through a preprogrammed 1D-LUT that maps the 7-bit input address (15), supplied as a pre-encoded decimal index, directly to the most appropriate SS. The 1D-LUT has 96 entries, derived from: $3! = 6$ EPC permutations, $2^3 = 8$ ESC combinations, and $2^1 = 2$ EMP states, resulting in $3 \cdot 8 \cdot 2 = 96$ total entries, where the remaining 32 entries of the potential $2^7 = 128$ LUT input addresses (15) are unused. Each entry corresponds to one of the 14 valid non-zero SSs (cf. Table I). Since the SE-VSI topology inherently links adjacent legs with phase currents, thus preventing simultaneous independent correction of multiple currents (cf. Section II), a strategic error-prioritized decision-making process is utilized. The SS selection process depends on the computed CEs and their characteristics, which are fully represented by the 7-bit input address (15). The LUT outputs the four switching functions of the SE-VSI, S_1, S_2, S_3, S_4 as an encoded decimal variable, which can be decoded into the eight SE-VSI's gate signals (cf. Figures 1 and 3). Due to the phase/leg interdependence, if one current is increased/decreased, any adjacent current can either be decreased/increased or preserved. For example, setting $S_1 = 1$ and $S_2 = 0$ applies $+V_{dc}$ to phase-A, increasing i_a . This leaves two options for governing i_b , either setting $S_3 = 1$ to decrease i_b by applying $-V_{dc}$ to phase-B, or setting $S_3 = 0$, preserving i_b through self-circulation by applying zero voltage to phase-B (cf. Fig. 1 and Table I).

TABLE III
SELECTED OPTIMAL SS LUT ENTRIES

Case#	EPO	EPC	ESC	EPM	SS	Impact
1	ACB	100	011	1	0110	$-i_a, +i_c$
2	ACB	100	001	0	0101	$-i_a, +i_b, -i_c, -i_0$
3	BAC	010	110	1	1101	$+i_b, -i_c$
4	BAC	010	110	0	1100	$+i_b, +i_0$
5	ABC	110	001	1	0110	$-i_a, +i_c$
6	ABC	110	001	0	0111	$-i_a, -i_0$
7	CAB	101	010	1	1001	$+i_a, -i_c$
8	CAB	101	110	0	1010	$+i_a, -i_b, +i_c, +i_0$

The SS selection process follows four fundamental rules that govern the LUT design:

1. **Prime Error Correction Priority:** The most critical error (prime error) as determined by the EPC is always corrected and never preserved. This ensures that the most significant control deviation receives immediate attention.
2. **Maximum Error Correction:** The selected SS attempts to correct as many CEs as possible. Secondary CE corrections are applied if they can be addressed without compromising the prime error correction.
3. **Non-Interference Principle:** Any secondary CE that cannot be immediately corrected is preserved through self-circulation of its corresponding current, thus ensuring a "do no harm" policy.
4. **Error Mode Predominance Selection:** When $F_{mp} = 1$ (DM predominance), the candidate SS subset consists of the 6 SSs that do not impact i_0 . Conversely, when $F_{mp} = 0$ (CM predominance), the candidate subset includes the 8 SSs that directly impact i_0 . This ensures optimal balancing between phase current regulation and ZSC suppression.

Since the full 96-entry LUT is too large to be listed entirely, Table III lists eight key LUT entries, while the full LUT is available online [21].

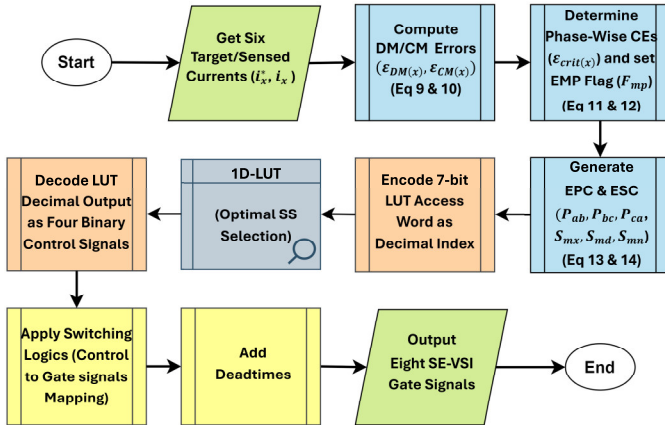


Fig. 3. Overall Implementation Flowchart of the main OEPC Control Framework with 1D-LUT-based Optimal Switching State Selection.

To illustrate the SS selection logic, consider four exemplary scenarios corresponding to cases #1 to #4 in Table III. In **Case #1**, the EPC = 100 (EPO = ACB) identifies $\varepsilon_{crit(a)}$ as the prime error, followed by $\varepsilon_{crit(c)}$ and $\varepsilon_{crit(b)}$. The ESC = 011 dictates that i_a must be decreased while i_c and i_b need to be increased if possible. Therefore, the LUT maps the input index to SS = 0110, which decreases i_a , increases i_c , and preserves i_b for a later control cycle. Note that since EPM = 1 (DM predominance), an i_0 -neutral SS is selected, allowing for a double CE correction. In **Case #2**, the EPC is unchanged (ACB), while ESC and EPM change to 001 and 0, respectively, dictating that i_a needs to be decreased, as well as i_c , if possible, while (if possible) i_b needs to be increased. This CM predominance case presents a special scenario where all three CEs can be corrected simultaneously by selecting the i_0 -impacting SS = 0101, thus decreeing i_0 by the expected 1/3-fold. In **Case #3**, the EPC = 010 (BAC) determines that $\varepsilon_{crit(b)}$

is the prime error, followed by $\varepsilon_{crit(a)}$, then $\varepsilon_{crit(c)}$. The ESC = 110 dictates that i_b must be increased, i_c should also be increased if possible, and i_a should decrease if possible. Due to the EMP = 1, a i_0 -neutral SS must be selected, narrowing it down to SS = 1101, thus correcting i_b and i_c , while preserving i_a . In **Case #4**, where the LUT input remains the same except for EMP = 0, thus indicating a CM predominance, the selected SS must now contribute to ZSC suppression. This time SS = 1100 is selected, where the impact on i_b and i_a is unchanged, but now i_a is preserved instead of decreased, thus reducing the number of CE corrections to only the prime error. This confirms that OEPC dynamically selects SSs based on error priority and their mode predominance by leveraging a 1D-LUT to efficiently select the optimal SS, ensuring all phase current errors and the ZSC error are always corrected within a few control cycles.

D. Overall OEPC Algorithm and Hardware Implementation Considerations.

The OEPC's overall framework is illustrated as a flowchart in Fig. 3, outlining the main procedural steps from current sensing to gate signal generation. The LUT-based optimized SS selection, which was detailed in Section III.C, is represented by a black-boxed subprocess. The OEPC algorithm depicted in Fig. 3 proceeds through the following main systematic stages:

1. **Error Generation:** Six current control errors are calculated - three DM errors (9) and three CM errors (10), representing deviations between reference and measured currents and quantifying all possible phase-wise correction demands.
2. **Error-Priority Processing:** Each phase's DM and CM errors are pairwise compared to select the dominant error (11). This reduces the total set to three CEs, one per phase, while simultaneously determining the EMP flag, which signals whether DM or CM correction should dominate the SS selection (12).
3. **Error Sorting and Classification:** The three CEs are sorted by magnitude, establishing a hierarchical error priority by generating the EPC (13) and the ECS (14), which encode the CEs magnitudes and signs, respectively. Together with the EMP flag, they form a 7-bit address uniquely identifying the system's error state (15).
4. **LUT-Based SS Selection:** The 7-bit address indexes into the 1D-LUT to determine the optimal SS by implementing the four fundamental rules described in Section III.C, ensuring that: (a) the prime error is always addressed; (b) maximum error correction is achieved where possible; (c) non-correctable errors are preserved through self-circulation; and (d) The balance between phase current regulation and ZSC suppression dynamically adapts based on the EMP.
5. **Gate Signal Generation:** The selected SS is decoded into eight complementary gate signals controlling the SE-VSI's four legs, which are refreshed each control cycle, at a self-adjusted f_{sw} with a bounded upper limit.

A key feature of OEPC architecture is its computational

efficiency driven by the direct lookup, thus eliminating the iterative calculations typical of 3D-SVM or predictive controllers. This allows for the support of hardware-efficient implementation. While validated on the TMS320F28335 MCU, the core LUT operation is reducible to a seven-input, four-output truth table. It can be directly mapped onto compact programmable logic arrays (PLAs), such as the GAL22V10. Paired with cost-effective MCUs (e.g., STM32F031) for error computation and speed loop handling, this MCU-PLA hybrid approach offers significant potential for further reducing system cost, latency, and power loss. Realization of the LUT as combinational logic enables fast response times, making OEPC suitable for applications demanding rapid control actions.

IV. VALIDATION AND PERFORMANCE EVALUATION

The proposed OEPC scheme was validated through simulations and real-time C-HIL experiments using an R-L load with 3rd-harmonic back-EMF, $e_0 = -E_0 \sin(3\omega t)$, (RLE test) and an OW-PMSM-based speed control drive (motor drive test). The motor drive simulation model implemented a linear VBR-based OW-PMSM in MATLAB Simulink [16], enhanced to include 3rd-harmonic ripple effects (cf. (1) - (5)). The OEPC algorithm was utilized as the current controller for both test cases. Real-time validation was conducted on a Typhoon HIL402 platform [22] interfaced with a DSP180 host board [23] running a TMS320F28335 MCU [24]. Real-time measurements and analysis were performed using a Keysight MSO-X3014T oscilloscope.



Fig. 4. Controller + Hardware in Loop (C-HIL) Experimental Setup.

A. OEPC Current Control Tests

While unsupported by OEPC - an inherently current control scheme, equivalent modulation indices of $\alpha\beta$ subspace, $m_{\alpha\beta}$, and CMV, m_{zs} , were computed for consistent comparison with voltage modulation-based methods, such as 3D-SVM [6] and CBPWM [8]. For the RLE load case, $m_{\alpha\beta}$ was computed from the fundamental current magnitude, I_f , while m_{zs} was set by controlling the 3rd harmonic back-EMF (E_0) as:

$$m_{\alpha\beta} = (I_f/V_{dc})\sqrt{R^2 + (\omega L)^2}, \quad m_{zs} = 3E_0/V_{dc}. \quad (16)$$

The mutual relationship between $m_{\alpha\beta}$ and m_{zs} is governed by the geometric modulation constraint, $m_{\alpha\beta} + (2/3)m_{zs} \leq 1$, [6], which defines the maximum allowable $m_{\alpha\beta}$ as a function of m_{zs} in the linear range, thus capturing the space sharing between DM and CM error correction feasibility in OEPC:

$$m_{\alpha\beta_max} = \min\left(1 - \frac{2}{3}m_{zs}, 1\right). \quad (17)$$

In the first set of tests, OEPC's sinusoidal current tracking, ZSC behavior, f_{sw} variability, and DC utilization capabilities were evaluated under different operating conditions using an RLE load, where E_0 and I_f^* are determined by the desired $m_{\alpha\beta}$ and m_{zs} values according to (16). R, L, V_{dc} , and f_{max} , were set to $8\Omega, 5mH, 60V$, and $5kHz$, respectively, to align with the 3D-SVM [6] RL load test conditions. Notably, OEPC operates as a self-oscillating scheme with a typical steady-state average f_{sw} , $\langle f_{sw} \rangle$ of approximately $0.75f_{max}$, which gives fixed- f_{sw} schemes like 3D-SVM, a potential advantage in current quality performance. Phase currents (i_x), ZSC (i_0) along with I_f and THD results are illustrated in Figures 5(a) to 5(c), while in Fig. 5(c) the cycle-to-cycle f_{sw} and $\langle f_{sw} \rangle$ are also revealed. The three figures depict the results of the following test cases:

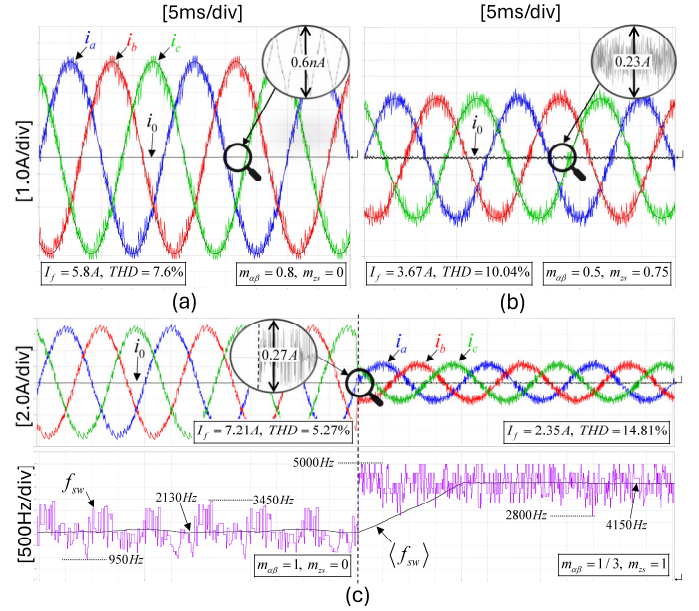


Fig. 5. OEPC Sinusoidal current tracking results under different conditions ($f_{max} = 5kHz$): (a) i_x, i_0 ($m_{\alpha\beta} = 0.8, m_{zs} = 0$), (b) i_x, i_0 ($m_{\alpha\beta} = 0.5, m_{zs} = 0.75$), (c) i_x, i_0, f_{sw} ($m_{\alpha\beta} = 1 \rightarrow 1/3, m_{zs} = 0 \rightarrow 1$).

(a) Ideal switches, linear load ($m_{\alpha\beta} = 0.8, m_{zs} = 0$): The absolute zero i_0 reveals the OEPC's capability to replicate a three-wire star-connected load behavior under CBPWM.

(b) Current quality at non-ideal switching ($m_{\alpha\beta} = 0.5, m_{zs} = 0.75$): Despite the advantage of 3D-SVM with a fixed $f_{sw} = 5kHz$ and a lower m_{zs} of 0.5, at an average f_{sw} of $3.75kHz$, OEPC maintains a competitive THD of 10% and a superior raw i_0 ripple of 0.23A, vs. 3D-SVM's 8.8% and 1.3A, respectively. Some minor ZSC ripples arise from deadtime effects.

(c) Transient behavior and maximum DC utilization ($m_{\alpha\beta} = 1 \rightarrow \frac{1}{3}, m_{zs} = 0 \rightarrow 1$): Simultaneously changing $m_{\alpha\beta}$ and m_{zs} between their boundary limits as per (16) while monitoring f_{sw} variation and $\langle f_{sw} \rangle$, highlights OEPC's robust transient behavior and efficient f_{sw} adaptivity, which reflects a general trait of comparator-based controllers, where lower reference magnitudes lead to higher f_{sw} due to rapid and aggressive correction/reversal cycles causing faster accumulation of error. Moreover, when f_{max} was increased from 5 to $50kHz$ for this test, current THD and ZSC ripple were both decreased from

14.81% and 2.35A to 1.42% and 0.023A, respectively, while maintaining maximal DC utilization, demonstrating OEPC's full ZSC suppression capability even at saturation boundary while maintaining minimal switching losses impact.

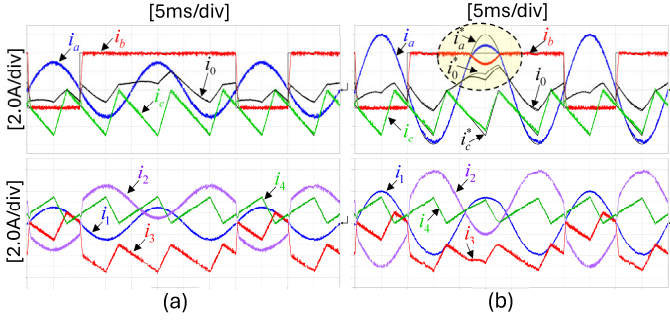


Fig. 6. OEPC response (phase and leg currents) under asymmetric references operation with transient overmodulation: (a) Maintained modulation constraints, (b) Periodically violated modulation constraints.

A second set of tests was conducted to assess the modulation boundary using three asymmetric reference currents: a sine wave, a rectangular wave, and a triangular wave, each characterized by distinct frequencies, amplitudes, duty cycles, phase shifts, and DC offsets. V_{dc} , L , and R , were consistent with the first test, while E_0 and f_{max} were set to 0 and 10 kHz, respectively. Fig. 6 displays the simulation results, which include the reference (i_x^*), the measured (i_x) currents, the ZSC (i_0), its effective reference (i_0^*), and the leg currents ($i_1 - i_4$). In Fig. 6(a), the 3.0A sinusoidal reference (i_a^*) remained within the modulation constraints, enabling accurate current tracking.

TABLE IV
OW-PMSM TEST CONFIGURATIONS (PMSM PARAMETERS)

#	V_{dc} [V]	L_d, L_q, L_0 [mH]	R_s [Ω]	p	J [gm^2]	λ_m [mWb]	K_m [%]
1	48	3.7, 5.0, 4.0	0.9	5	1.0	80	2.5
2	60	1.5, 1.8, 2.2	0.4	5	1.0	80	2.5
3	60	3.7, 5.0, 4.0	0.9	5	1.0	22	2.5
4	200	4.0, 7.0, 6.0	2.0	4	2.0	175	5.0

In Fig. 6(b), doubling i_x^* to 6.0A caused it to exceed the boundary periodically. During those overmodulation intervals, OEPC displayed a unique self-regulating behavior. Instead of saturating, it dynamically distributed control priority among the competing errors, causing temporary current underflows and full recovery once the feasible modulation was restored. These results illustrate OEPC's robustness under extreme asymmetry and its ability to handle transient infeasibility gracefully. While conventional overmodulation support could be incorporated for applications requiring saturation (e.g., via logic flags or LUT switching), OEPC's default behavior inherently prevents harmonic distortion and maintains control predictability. Moreover, across all test scenarios, the feasible modulation range and the DC utilization have shown compliance with the traditional modulation constraint in (16) and (17), confirming OEPC's adherence to SE-VSI modulation limits while achieving robust current control and superior current quality.

B. OEPC Motor Drive Tests

OEPC-based SEW-PMSM tests were conducted across four configurations, which are detailed in Table IV. Configurations

#1- #3 fully align with those from [4], [5], and [6], respectively, while #4 is a higher-power OW-PMSM drive with a pronounced 3rd harmonic flux component. Table V lists the reported results from [4] – [6] at $f_{sw} = f_{max} = 20kHz$, alongside OEPC results under identical conditions (superior results are bolded). Table V reveals that OEPC improves ZSC ripples and achieves competitive current THD.

TABLE V
REPORTED PMSM DRIVE RESULTS VS. OEPC-BASED REPLICATIONS

Methodology	Test Cfg #	Speed [rpm]	T_m [Nm]	I_f [A]	Δi_0 [A]	THD [%]
MVMPC [4]	1	100	0.4	1.5	0.20	6.10
OEPC					0.17	6.84
MVMPC [4]	1	1000	2.3	8.0	1.60	7.51
OEPC					0.24	1.69
AZSMPC [5]	2	100	0.5	3.0	0.18	5.02
OEPC					0.14	5.75
AZSMPC [5]	2	500	0.5	3.1	0.20	5.81
OEPC					0.14	6.54
3D-SVM [6]	3	500	1.0	3.8	1.05	-
OEPC					0.24	4.31
3D-SVM [6]	3	1000	1.5	5.0	1.20	-
OEPC					0.27	3.08

It should be noted that discrepancies were observed between reported torque levels and implied current magnitudes in [4] and [6], where results could not be replicated using the stated machine models and test conditions unless a field weakening condition of $i_d^* = -2i_q^*$ was enforced, deviating from the typical $i_d^* = 0$ in standard FOC [14]. This was the only way to align simulation results with the reported 2x fold inflated current amplitudes in [4] and [6] (i.e., 8A at 2.3Nm). While this behavior may stem from machine-related constraints or reflect a design choice, it was not explicitly addressed in [4] and [6].

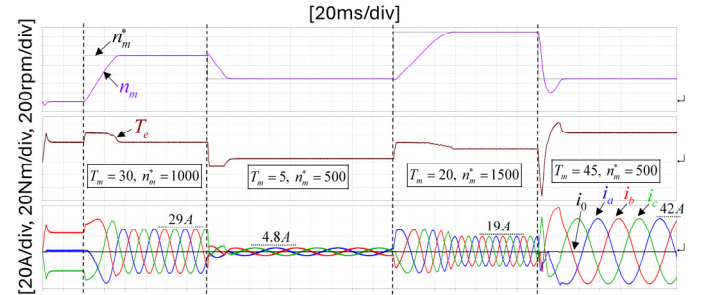


Fig. 7. Dynamic Response of the SEW-OEPC-Based PMSM Drive Under Speed and Load Variations.

Fig. 7 presents the dynamic response of the SEW-OEPC-driven OW-PMSM drive, corresponding to configuration #4 in Table IV (240VAC, 30Nm, 1000RPM rated) at $f_{max} = 20kHz$, under sequential speed and load torque variations. The test sequence progresses through multiple steady-state operating points (T_m, n_m) including low-speed light-load (500RPM, 5Nm), high-speed medium-load (1500 RPM, 20Nm), and abrupt transitions such as acceleration from standstill to 1000 RPM at full load and deceleration from 1500 RPM to 500RPM while simultaneously increasing the load torque to 45Nm. Throughout these transitions, the drive demonstrates excellent reference tracking, disturbance rejection, and current regulation. Torque ripple remains within $\pm 0.5Nm$ under all

conditions, with no current transient oscillations observed, highlighting the OEPC's robust control performance.

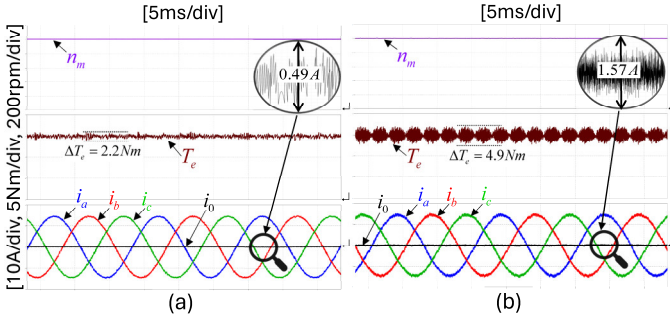


Fig. 8. Steady-State Speed, Torque, and phase Current responses of SE-VSI-driven OW-PMSM under high 3rd harmonic flux conditions ($K_m = 20\%$) and 6% overmodulation: (a) OEPC, (b) CBPWM [8].

Fig. 8 compares the steady-state responses of SE-VSI-driven OW-PMSM (configuration #4) using OEPC and CBPWM [8] under prominent 3rd harmonic back-EMF conditions, with K_m increased to 20% at rated speed and 50% load ($n_m = 1000\text{RPM}$, $T_m = 15\text{Nm}$), resulting in modulation indices of $m_{\alpha\beta} = 0.62$ and $m_{zs} = 0.57$, leading to 6% overmodulation ($m_{\alpha\beta} + 1.5m_{zs} = 1.06$), as per (17). While such a high K_m is impractical (typically $<5\%$), it ensures quite a challenging scenario for ZSC and torque ripple suppression.

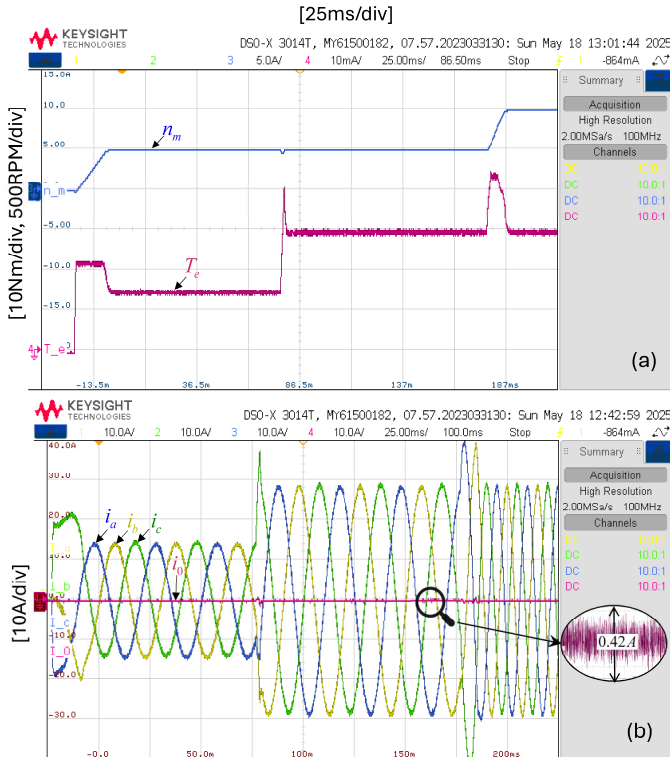


Fig. 9. Real-time experimental results of OEPC-based motor drive control (C-HIL with TMS320F28335 MCU) and emulated OW-PMSM: (a) Speed and electrical torque response under sequential speed and load changes; (b) Current (phase + ZSC) responses.

As Fig. 8(a) shows, OEPC ($f_{\text{max}} = 20\text{kHz}$) achieves low torque and ZSC ripples of 2.2Nm and 0.49A with 2.26% phase current THD. Conversely, the CBPWM results in Fig. 8(b)

show higher ripples (4.9Nm, 1.57A) and 3.12% current THD. CBPWM also exhibits a strong 6th harmonic component in torque ripple from its linear ZSC compensator, causing degraded performance near overmodulation [17]. Despite operating at lower f_{sw} ($f_{\text{sw}} \approx 16\text{kHz}$), OEPC reduces switching losses by $\sim 20\%$ compared to CBPWM thanks to its adaptive f_{sw} utilization. These results confirm OEPC's superior ZSC and torque ripple suppression, improved current quality and reduced switching losses even under harsh conditions.

Fig. 9 presents real-time C-HIL experimental results of OEPC-based motor drive control. The test employs an emulated OW-PMSM (configuration #4) with a 10-bit PPR and a 20kHz BiSS-based resolver carrier, representative of real-world applications. Starting at 500RPM/15Nm, the speed reference was stepped to 1000RPM, followed by a load torque increase to 30Nm (via the HIL SCADA interface), while capturing the triggered events' responses. Fig. 9(a) shows the speed and torque responses, while Fig. 9(b) shows the phase currents and ZSC, all demonstrating excellent dynamic performance and regulation, validating the simulation predictions. Additionally, complete control execution time measured from ADC trigger of speed sensor (encoder) to GPIO update at 40kHz sampling ($f_{\text{max}} = 20\text{kHz}$) revealed that the OEPC cycle, including FOC, completed in $1.75\mu\text{s}$ on the TMS320F28335 (150 MIPS, single core). In contrast, the 3D-SVM scheme reported in [6] required $0.43\mu\text{s}$ on the faster dSPACE-1202 platform (2GHz, dual-core), extrapolating to $\sim 11\mu\text{s}$ if ported to the TMS - over 10-fold slower than OEPC. At steady-state full load ($n_m = 1000\text{RPM}$, $T_m = 30\text{Nm}$), the measured power efficiency for the SE-VSI reached 94% when accounting for parasitics, IGBT losses, and deadtime. FFT analysis revealed a current THD of 1.84% and a 0.42A ZSC ripple, closely matching simulated results. These findings confirm OEPC's practical viability, robustness, and efficiency on cost-effective digital hardware.

V. CONCLUSION

This paper introduced the Optimal Error-Priority Control (OEPC) framework for series-end VSI-fed PMSM drives, providing direct ZSC suppression in ABC coordinates without modulation synthesis or duty cycle averaging. OEPC employs a compact 1D-LUT and variable-frequency phase-wise approach to achieve efficient ZSC control and reduced switching losses while maintaining high current quality and torque stability. Simulations and C-HIL experiments fully validated OEPC's performance under non-ideal conditions, such as OW-PMSMs with significant 3rd harmonic content, confirming robust dynamic behavior with sub- $2\mu\text{s}$ execution time on the TMS320F28335, outperforming existing methods in computational efficiency. The OEPC's simplicity makes it ideal for embedded implementation in industrial drives, with potential extensions to multiphase SE-VSI configurations for high-performance, cost-sensitive motor control systems.

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